Health and Safety at Work

'Control of Substances Hazardous to Health Regulations (CoSHH)'

Learning aims

You should be able to:

Understand the purpose and content of the **‘Control of Substances Hazardous to Health Regulations’ (COSHH) 2002**

What is a hazardous substance under the Regulations?

COSHH covers substances that are hazardous to health. Substances can take many forms and include:

- *Chemicals – or products containing chemicals*
- *fumes*
- *dusts*
- *vapours*
- *mists*
- *nanotechnology*
- *gases and asphyxiating gases and*
- *biological agents (germs).*



If the ***packaging*** has any of the hazard symbols then it is classed as a hazardous substance.

What are ***NOT*** hazardous substances under CoSHH?

These have their own regulations:

- *Lead*
- *Asbestos*
- *Radio-active substances*

What must an *employer* do?

Eliminate or reduce risks from hazardous substances

This is achieved by:

- ***Risk assessment***
- ***Control measures***
- ***Monitoring health & exposure if necessary***
- ***Information, training and supervision***



What must an ***employee*** do?

- Take reasonable care of our own safety and not endanger others
- Cooperate with our employer
- Make full and proper use of control measure



*Just **reiterating** our duties under HSAWA 1974*

Risk Assessment for CoSHH

Step 1:
Risk Assessment for CoSHH

Identify the hazardous substance(s)

e.g…

- *Adhesives*
- *Paints*
- *Polishes*
- *Machining Coolants*
- *Oils/Grease*
- *Solder Fuses/Flux*
- *MDF Dust*
- *PCB Developer/Etchant*
- *Cleaning Fluids*















*Look at the **'labels'** on the packaging.*

Step 1:
Risk Assessment for CoSHH

New International Hazard Symbols



Danger



Flammable



Oxidiser

Step 1:
Risk Assessment for CoSHH

New International Hazard Symbols



Explosive



Corrosive



Compressed or liquefied gas

Step 1:
Risk Assessment for CoSHH

New International Hazard Symbols



Aquatic Toxicity



Warning



Sensitiser, carcinogen, mutagen or teratogen

OTHER SYMBOLS YOU MIGHT SEE – WHAT DO THEY MEAN?























EXPLOSIVE



FLAMMABLE



HIGHLY FLAMMABLE



OXIDISING



CORROSIVE



TOXIC



VERY TOXIC



HARMFUL TO THE ENVIRONMENT



IRRITANT / HARMFUL



BIOLOGICAL AGENT

Step 1:
Other sources of information



- **Use available data e.g...**
  - Data sheets & labels
  - On-line databases

Step 1
Decide ***who*** is at risk and how

- *Staff*
- *Students*
- *Others*



Consider their current training (if any) and their background

Step 1:
Decide who is at risk and ***how***

- *Inhalation*
- *Skin or eye contamination*
- *Ingestion*
- *Injection*



Step 1:
Evaluate the Risk

Factors to Consider:

- *Toxicity*
- *Form (gas, spray, dust, liquid, solid)*
- *Solubility*
- *Amount (weight &/or volume)*
- *Nature of the operation*
- *Length of exposure*
- *Number of people involved*



Step 2:
Decide on Control Measures

- Control exposure in **proportion to risk** by using the hierarchy of controls.

- Personal protective equipment should be a last resort as the prime means of control



Step 3:
Use of Control Measures (1)



- **Replace substance with a *safer* alternative**

- E.g. use a lower hazard disinfectant rather than bleach (irritant) if it will do the job adequately

Step 3:
Use of Control Measures (2)

 Use the material in a safer form:

e.g…

- Use water-based paint instead of solvent-based paint



- Buy hazardous materials in pre-weighed sachets rather than having to measure & make-up from bulk quantities

Step 3:
Use of Control Measures (3)

Control the operation

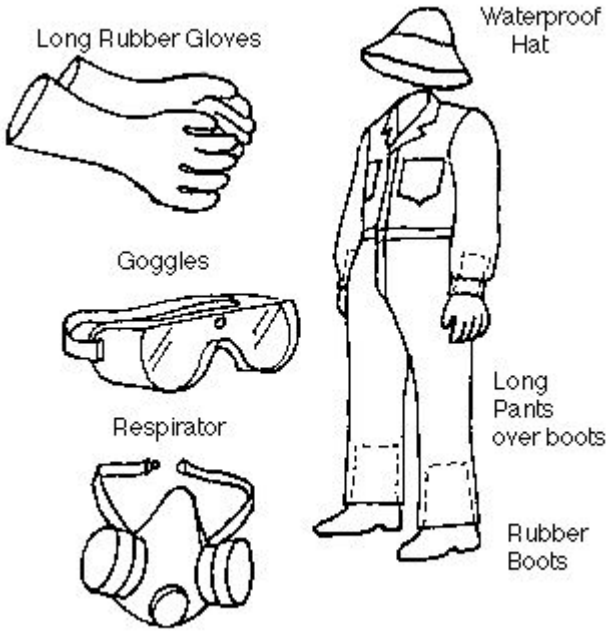
e.g…

- Isolate the work
- Control at source
  - *Local exhaust ventilation*
- Reduce the number of workers
- Reduce the frequency



Step 3:
Use of Control Measures (4)

- **Personal protective equipment as a last resort:**
  - *Protects only the worker & not others in the room*
  - *Training & maintenance required*
  - *Often not very comfortable*



Step 3:
Use of Control Measures (5)

Good workshop techniques are vital

e.g…

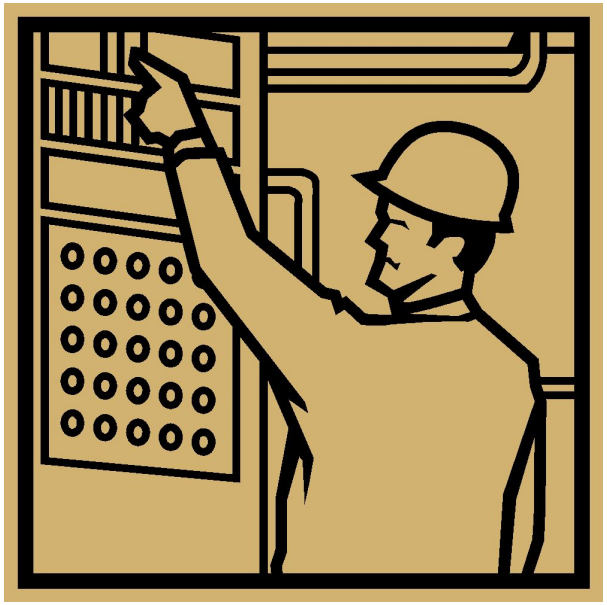
- Labelling
- Correct substance storage
- Warning signs where appropriate
- Cleanliness & tidiness
- Correct waste disposal





Step 4:
Maintenance of control measures

- Workshop equipment must be kept in good repair & working properly
- Regular simple checks on airflow on LEV equipment



- LEV must have engineering checks every 14 months
- Records kept for 5 years

Step 5:
Derive safe working procedures

- Preparation phase (e.g. weighing out)
- The process itself
- Safe waste disposal
- Emergency procedures
  - Spillage
  - Fire
  - First aid



Step 6:
Check and review:

- Are the control measures adequate?
- Are they working correctly?
- Is everyone aware of how to use them?

